Good Soil

Hands-on STEM, Arts & Skills Academy — Parent Curriculum Guide

25-Class Master STEM & Robotics Curriculum
Junior Cohort (Ages 7–10) & Senior Cohort (Ages 10–14+)
50 Total Class Hours · 25 Weekend Sessions (2 hrs / week)
Website: **goodsoil.in** · New Delhi, India

TOTAL CLASSES	TAKE-HOME KITS	STUDIO GROUP LABS	MILESTONE CHALLENGES
25 Classes	**10 Builds to Keep**	**10 Team Challenges**	**4 + 1 Grand Expo**
50 Hands-on Class Hours	Constructed, Tested & Kept	Robotics, IoT, Aero & AI Labs	Non-Paper Creative Derbies

> **"Every child is a seed. We're the good soil."**
>
> Good Soil was born from over a decade of hands-on aeromodelling workshops with Airmodelcrafts (AMC). We learned that what truly ignites young minds is not memorizing textbook diagrams — it's the thrill of building something with their own hands, troubleshooting when it doesn't work the first time, and seeing it take flight.

The 4-Phase Learning Pathway

PROGRESSIVE SKILL BUILDING

PHASE 1 · CLASSES 01–06
Spark & Circuits
Electric loops, battery polarity, switches, and breadboarding. Concludes with Milestone Challenge 1: Island Escape.

PHASE 2 · CLASSES 07–12
Sensing & Mobile Bots
Light sensors (LDR), buzzers, mechanical gearboxes, and building a rolling mobile obstacle rover. Milestone 2: Carnival Derby.

PHASE 3 · CLASSES 13–18
Logic, Code & Flight
Microcontrollers, block coding, sensor feedback, and aeromodelling flight dynamics. Milestone 3: Sky Patrol.

PHASE 4 · CLASSES 19–25
Kinematics, Drones & Expo
Bionic linkages, hydraulic arms, mini-drone avionics, solar rovers, and the Grand Capstone Expo with visiting parents.

Inside a 120-Minute Good Soil Class

STRUCTURE & DISCIPLINE

1. Discovery & Starter (25 min)
Short visual story and tactile exploration of loose parts. Kids formulate a hypothesis before touching any assembly steps.

2. Guided Project Build (55 min)
Step-by-step physical construction with real tools. Safe wiring, mechanical alignment, and mentor-guided calibration.

3. Test, Flight & Passport (40 min)
Bench testing, timed obstacle trials, and logging measurements in their personal student Good Soil STEM Passport.

Our Core Educational Commitments

THE GOOD SOIL PROMISE

🎁 10 Take-Home Builds
Kids permanently keep 10 complete working STEM builds to share pride at home.

🚫 100% Non-Paper Tests
Zero written tests — mastery is proven through live obstacle challenges and derbies.

🛡️ Safety & 1:8 Mentorship
Low-voltage child-safe DC circuits, eye protection, and intimate 1:8 supervision.

🎓 STEM Passport & Expo
Milestone badge stamps and a celebratory public Capstone Expo for families.

Junior Cohort Syllabus (Ages 7–10 / Grades 2–4)

Part 1: Circuits, Motors & Mobile Bots (Classes 01–12)

Track Rhythm: 2 Hours / Session · Weekly Schedule
Focus: Tactile First-Principles, Breadboarding & Mobility

Cls	Ph	Activity & Project Build	Format	Key Concepts & Real-World Skills Mastered
01	P1	**Human vs. Robot Explorer Station** Module 1: Meet the Robots	Studio Group Lab	Tools vs. autonomous machines; how sensors act as robotic sensory organs.
02	P1	**10-in-1 Electricity & Switch Lab** Module 2: Circuit Pathways & Electric Flow	Studio Group Lab	Closed loops, battery polarity (+/-), knife switches, incandescent bulbs & high-intensity LEDs.
03	P1	**5 Organs of a Robot Dissection** Module 3: Simple Robot Architecture	Studio Group Lab	Hands-on dissection of the 5 robot pillars: Power, Sensor, Controller, Motor, and Chassis.
04	P1	**Signal Logic Traffic Light Circuit** Module 4: Solderless Breadboard Intro	Take-Home Kit #1	Constructing a 3-colour traffic signal; switch logic, resistor networks, breadboard rails.
05	P1	**Aerodynamic Propeller Wind Car** Module 5: Motorized Aerodynamic Thrust	Take-Home Kit #2	DC motor torque, propeller pitch direction, conversion of electrical energy into wind thrust.
06	P1	**Milestone 1: Island Escape Challenge** Non-Paper Team Build Challenge	Challenge & Badge	**Teams combine wind cars and circuit signals to solve an obstacle challenge.** ***Badge: Junior Explorer***
07	P2	**Wooden Glow Night Lamp Circuit** Module 6: Light Sensors & Automation	Take-Home Kit #3	Light-Dependent Resistors (LDR), automatic night-time activation, sensor-triggered circuits.
08	P2	**Telegraph & Electromagnetic Buzzer Lab** Module 7: Sound & Magnetism	Take-Home Kit #4	Electromagnetic coils, Morse code communication, magnetic reed switches, sound signaling.
09	P2	**Reversible Railway Gate Machine** Module 8: Motor Reversal & Limit Switches	Studio Group Lab	Push-button polarity flipping, automated barrier gate motion, forward/reverse gearing.
10	P2	**Mechanical Gearbox Carousel** Module 9: Transmission & Torque	Studio Group Lab	Spur gears, driver vs driven gears, mechanical advantage, trading rotational speed for torque.
11	P2	**Junior 2-Wheel Mobile Rover** Module 10: Complete Robot Integration	Take-Home Kit #5	Integrating chassis, dual motors, power switch, and differential wheel steering.
12	P2	**Milestone 2: STEAM Carnival Derby** Non-Paper Team Build Challenge	Challenge & Badge	**Fairground arena build combining gear machines and rovers.** ***Badge: Master Crafter***

Junior Cohort Syllabus (Ages 7–10 / Grades 2–4)

Part 2: Microcontrollers, Aeromodelling & Grand Expo (Classes 13–25)

Track Rhythm: 2 Hours / Session · Weekly Schedule
Focus: Block Code, Flight Physics, Kinematics & Graduation

Cls	Ph	Activity & Project Build	Format	Key Concepts & Real-World Skills Mastered
13	P3	**Microcontroller Block Brain First Blink** Module 11: Introduction to Controllers	🤝 Studio Group Lab	Robot microcontrollers, GPIO output pins, drag-and-drop block coding, LED matrix animations.
14	P3	**Interactive Emotion & Tone Display** Module 12: Digital Sensors & Feedback	🤝 Studio Group Lab	Button event listeners, animated OLED facial emotions, piezo frequency musical synthesis.
15	P3	**Smart Multi-Sensor Security Alarm** Module 13: Integrated Sensor Robotics	🎁 Take-Home Kit #6	Obstacle proximity sensors, audio-visual alarm logic, home security automation.
16	P3	**Looping Light Patrol & Dancing Bot** Module 14: Loops & Repetition	🤝 Studio Group Lab	While/repeat programming loops, sequence timing, autonomous cyclic motor behaviors.
17	P3	**Rubber-Band Aerodynamic Glider** Aeromodelling Track I: Flight Forces	🤝 Studio Group Lab	The 4 Forces of Flight (Lift, Weight, Thrust, Drag); balsa airfoil assembly, test flight trim.
18	P3	⏳ **Milestone 3: Sky Patrol Beacon Derby** Non-Paper Team Build Challenge	🎯 Challenge & Badge	**Target landing accuracy and Morse code communication trials.** ***Badge: Flight Cadet***
19	P4	**Walking Bionic Creature Linkage** Module 15: Kinematics & Linkages	🎁 Take-Home Kit #7	4-bar linkage mechanisms, converting rotational motor spin into rhythmic walking gait.
20	P4	**Catapult & Trajectory Launcher** Mechanical Physics & Levers	🎁 Take-Home Kit #8	Class 1 levers, fulcrum positioning, converting elastic potential energy to kinetic trajectory.
21	P4	**Balsa Chuck Glider Build & Tune** Aeromodelling Track II: Stability	🎁 Take-Home Kit #9	Center of Gravity (CG) balancing with nose ballast, dihedral wing angle, pitch stability.
22	P4	**Mini-Drone Flight Discovery Station** Aeromodelling Track III: Drone Physics	🤝 Studio Group Lab	Quadcopter aerodynamics, counter-rotating propellers, roll/pitch/yaw pilot flight control.
23	P4	**Solar-Powered Eco Rover Car** Clean Energy & Final Prep	🎁 Take-Home Kit #10	Photovoltaic cells, renewable solar energy conversion, capstone project integration.
24	P4	⏳ **Milestone 4: Future City Arena** Capstone Integration Trial	🎯 Challenge & Badge	**Integrating walking robots, rovers, and gliders in an eco-city pitch.** ***Badge: Young Innovator***
25	P4	🎓 **Good Soil Junior STEM Expo & Graduation** **Grand Public Showcase Day for Parents**	🌟 Expo & Ceremony	**Live glider flights, obstacle arena runs, portfolio defense, medal ceremony & graduation.**

Senior Cohort Syllabus (Ages 10–14+ / Grades 5–9)

Part 1: Schematics, Transistors & 4WD Mobility (Classes 01–12)

Track Rhythm: 2 Hours / Session · Weekly Schedule
Focus: Circuit Design, Solid-State Logic & Kinematics

Cls	Ph	Activity & Project Build	Format	Key Concepts & Real-World Skills Mastered
01	P1	**Autonomous Systems & Degrees of Freedom** Module 1: Robotics Kinematics	🤝 Studio Group Lab	Degrees of Freedom (DoF), kinematic chains, Cartesian vs articulated arm geometries.
02	P1	**Pseudocode, Logic & Flowcharts** Module 2: Algorithmic Architecture	🤝 Studio Group Lab	IF-THEN-ELSE decision trees, state transition diagrams for automated robotics.
03	P1	**Hardware Schematics & Routing** Module 3: Electronics System Architecture	🤝 Studio Group Lab	Reading schematic diagrams, analog vs digital signal buses, IC datasheet analysis.
04	P1	**Advanced Prototyping Breadboard Lab** Module 4: Solderless Circuit Design	🎁 Take-Home Kit #1	Mastering breadboard tie-points, dual 5V/3.3V power rails, signal wire organization.
05	P1	**Multi-Voltage Power Supply Lab** Module 5: Voltage Regulation & Testing	🎁 Take-Home Kit #2	Ohm's Law (V=IR), current loops, testing voltage drop with digital multimeters.
06	P1	🏆 **Milestone 1: Diagnostics Hackathon** Non-Paper Team Build Challenge	🎯 Challenge & Badge	**Fault diagnosis, multimeter continuity testing, circuit isolation.** ***Badge: Apprentice Engineer***
07	P2	**Tilt & Vibration Motion Sense Circuit** Module 6: Mechanical Sensor Triggers	🎁 Take-Home Kit #3	Normally-Open vs Normally-Closed switch contacts, mechanical tilt sensing, latching triggers.
08	P2	**Transistor Rain Alarm & Audio Stage** Module 7: Solid-State Transistor Logic	🎁 Take-Home Kit #4	NPN bipolar junction transistors (BJT), using transistors as solid-state switches, audio alerts.
09	P2	**Relay Logic & H-Bridge Motor Control** Module 8: High-Current Switching	🤝 Studio Group Lab	Electromechanical relays, DPDT switches, bidirectional H-bridge DC motor control.
10	P2	**Speed vs. Torque Gear Ratio Dynamometer** Module 9: Mechanical Kinematics	🤝 Studio Group Lab	Compound gear trains, spur vs worm gearing, calculating gear ratios (N1/N2) and stall torque.
11	P2	**4WD Autonomous Robot Chassis** Module 10: Complete Mobile Platform	🎁 Take-Home Kit #5	Dual-axle gearbox driving 4 rubber tires, weight distribution, motor alignment.
12	P2	🏆 **Milestone 2: Rescue Rover Canyon Race** Non-Paper Team Build Challenge	🎯 Challenge & Badge	**Teams drive 4WD rovers over incline ramps and rough terrain.** ***Badge: Systems Builder***

Senior Cohort Syllabus (Ages 10–14+ / Grades 5–9)

Part 2: Embedded C++, Avionics, Hydraulics & Capstone (Classes 13–25)

Track Rhythm: 2 Hours / Session · Weekly Schedule
Focus: Microcontrollers, Flight Dynamics, IoT & Capstone Defense

Cls	Ph	Activity & Project Build	Format	Key Concepts & Real-World Skills Mastered
13	P3	**Microcontroller Architecture & GPIO Code** Module 11: Embedded Systems Intro	🤝 Studio Group Lab	Microcontroller architecture, digital GPIO, uploading C++/block firmware routines.
14	P3	**Analog Read & PWM Motor Speed Control** Module 12: Pulse Width Modulation	🤝 Studio Group Lab	Pulse-Width Modulation (PWM), duty cycle variation, analogRead() for precision throttle.
15	P3	**Infrared Proximity Radar & Alarm** Module 13: Sensor Array Processing	🎁 Take-Home Kit #6	Multi-stage IR distance detection, visual LED proximity sweep, audible frequency alert.
16	P3	**Finite State Machines & Edge Avoidance** Module 14: Non-Blocking Automation	🤝 Studio Group Lab	State machines, edge detection sensors, non-blocking timing algorithms for roving bots.
17	P3	**Balsa High-Aspect Rubber Monoplane** Aeromodelling Track I: Wing Physics	🎁 Take-Home Kit #7	Cambered airfoils, angle of attack, aspect ratio, geared rubber motor endurance flights.
18	P3	⏳ **Milestone 3: Flight Derby & Radar Arena** Non-Paper Team Build Challenge	🎯 Challenge & Badge	**Timed flight endurance trials and obstacle avoidance arena runs.** ***Badge: Avionics Specialist***
19	P4	**Kinematic Linkages & JCB Excavator Arm** Module 15: Mechanical Engineering	🤝 Studio Group Lab	Grashof's criteria, 4-bar linkage mechanisms, slider-crank excavator arm physics.
20	P4	**Hydraulic Gripper Claw Robot Arm** Fluid Power & Hydraulics	🎁 Take-Home Kit #8	Pascal's Law (P=F/A), incompressible fluid force transmission, multi-axis master/slave syringes.
21	P4	**Electric Motorized Power Aircraft** Aeromodelling Track II: Powered Flight	🎁 Take-Home Kit #9	Thrust-to-Weight ratio (T/W), coreless electric motor, battery discharge, powered climb.
22	P4	**Quadcopter Flight & Telemetry Lab** Aeromodelling Track III: Drone Avionics	🤝 Studio Group Lab	Counter-rotating rotors, gyroscope balance, 3-axis flight PID principles, live telemetry.
23	P4	**Smart Connected IoT Weather Station** Connected Devices & Telemetry	🎁 Take-Home Kit #10	Environmental sensors, wireless data packet broadcasting, real-time IoT dashboard graphs.
24	P4	⏳ **Milestone 4: Shark Tank & Capstone Defense** Capstone Design Presentation	🎯 Challenge & Badge	**Teams pitch customized prototypes, explain schematics, and defend design trade-offs.** ***Badge: Master Innovator***
25	P4	🎓 **Senior Technology Symposium & Graduation** Grand Public Showcase Day for Parents	☀️ Expo & Ceremony	**Live robot course challenge, powered flight demonstrations, graduation honors & portfolio handoff.**

Student Experience & Parent Guide

Take-Home Kits, Workshop Safety & Admissions

Batches & Admissions: Open Year-Round
Experience Center: New Delhi, India

The 10 Take-Home Kits Every Student Keeps Forever

HANDS-ON PRIDE

Junior Cohort (Ages 7–10) Take-Home Builds:

1. **Traffic Light Signal Kit:** 3-LED manual switch breadboard circuit.
2. **Propeller Wind Car:** Motorized thrust vehicle with aerofoil blade.
3. **Night Lamp Glow Circuit:** Automated LDR darkness-detecting light.
4. **Telegraph & Buzzer Station:** Electromagnetic Morse sound key.
5. **2-Wheel Obstacle Rover:** Complete battery-powered rolling robot.
6. **Multi-Sensor Robotics Box:** Proximity security alarm detector.
7. **Bionic Walking Creature:** 4-bar linkage rhythmic walking automaton.
8. **Catapult Trajectory Launcher:** Kinetic lever and target system.
9. **Balsa Chuck Glider:** Aerodynamic balsa plane with CG trim balance.
10. **Solar-Powered Eco Rover:** Photovoltaic green energy vehicle.

Senior Cohort (Ages 10–14+) Take-Home Builds:

1. **Prototyping Breadboard Station:** Dual 5V/3.3V power module rig.
2. **Multi-Voltage Power Supply:** Ohm's law load testing station.
3. **Motion Sense Circuit:** Mechanical tilt & vibration alarm system.
4. **Transistor Piezo Rain Alarm:** BJT solid-state electronic switch.
5. **4WD Autonomous Chassis:** 4-motor high-traction mobile robot.
6. **Infrared Reversing Radar:** Multi-LED distance indicator and alarm.
7. **Balsa Monoplane:** High-aspect balsa skeleton with rubber gearbox.
8. **Hydraulic Gripper Robot Arm:** Pascal fluid claw mechanics.
9. **Motorized Electric Aircraft:** Coreless high-RPM powered climber.
10. **Smart Connected IoT Node:** Telemetry dashboard and smart rover.

Frequently Asked Questions & Studio Policies

SAFETY & STRUCTURE

Do kids need prior coding or robotics experience?

No prior background is required! Our curriculum begins with tactile first-principles before introducing microcontrollers. Every build is mentor-guided step-by-step.

What are the batch schedules & format options?

We offer 3 flexible schedules: **Weekend Classes** (2 hrs every Sat or Sun), **2-Day Workshops** (3 hrs/day on holidays), and **5-Day Holiday Camps** (4 hrs/day immersion).

How are safety and tools handled in the workshop?

All circuits use low-voltage DC batteries (3V to 9V) — no mains electricity is ever handled. Students receive dedicated safety briefings with eye protection under 1:8 supervision.

What happens if a child misses a weekend session?

We provide dedicated 45-minute catch-up clinics before the start of the next weekend's class, ensuring no student falls behind or misses assembling their take-home kit.

How to Reserve a Seat

JOIN GOOD SOIL

Ready to sow the seed of curious making?

Visit **goodsoil.in/classes** to view upcoming batch dates or reserve a seat directly via WhatsApp.

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